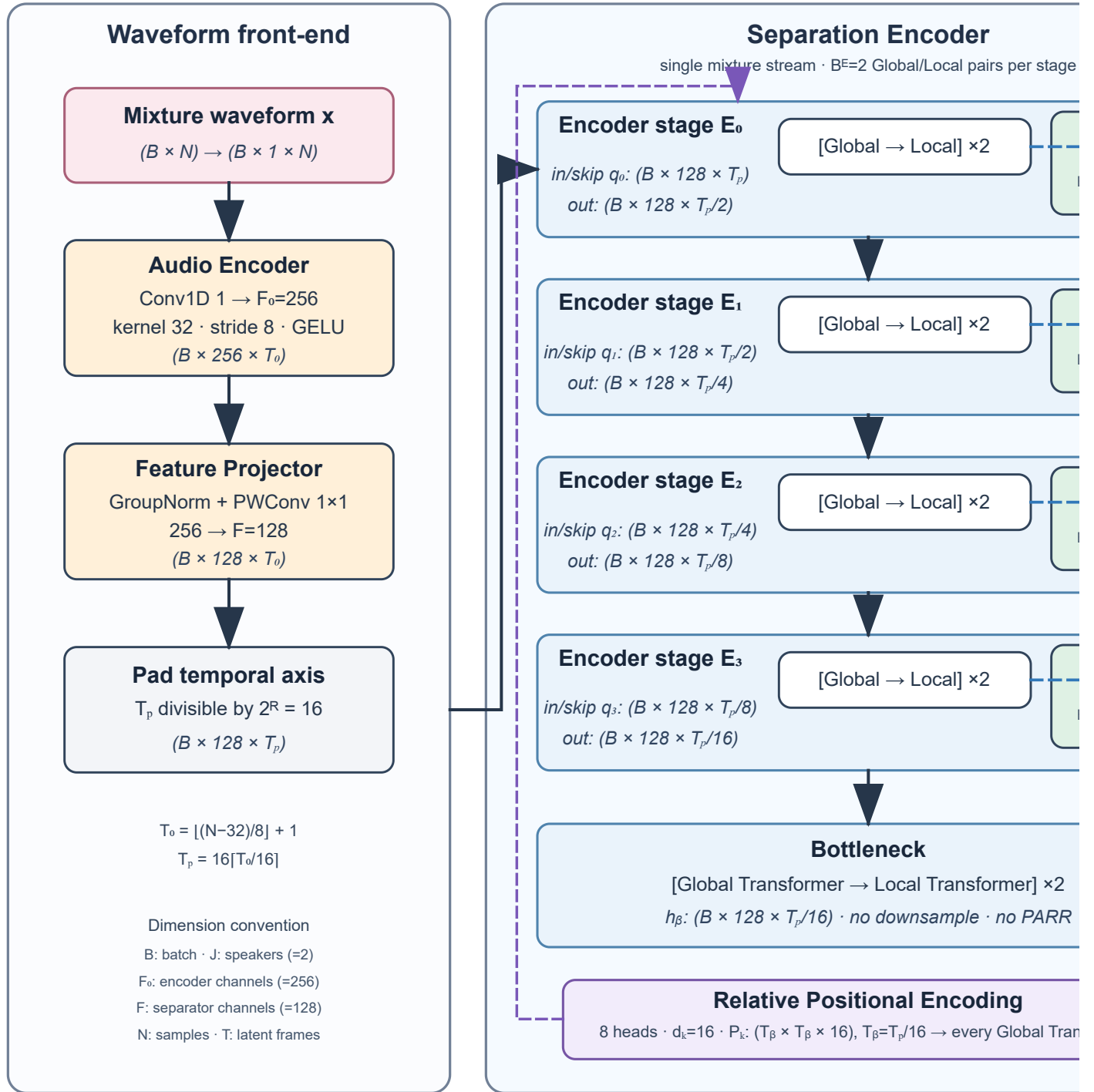


A. End-to-end architecture and multi-scale separator



B. PARR_r detail — exact forward path

Input/output use channel-first convention. $C_\beta=64$; the zero-initialized controller produces a channel-temporal gate.

